Computing & Programming Guide: Using Python and R on Datahub

Most data analysis and econometrics tasks can be performed equally well across different programming languages, but you should choose one and get accustomed to it. Please remember that you should NOT rely on a spreadsheet program for these tasks.

While STATA has been used historically in economics courses, **please note that we no longer have access to STATA, even through the Citrix server.** Because of this, we strongly encourage you to use **Python or R**, both of which are completely free programming languages.

Additional Help

Berkeley's D-Lab homepage is dlab.berkeley.edu/home. D-Lab has many tutorials, workshops, and help materials for Python, R, and data analysis. You can also request a free one-on-one consultation if you want help getting started with Datahub, Python, R, or setting up tools on your own computer.

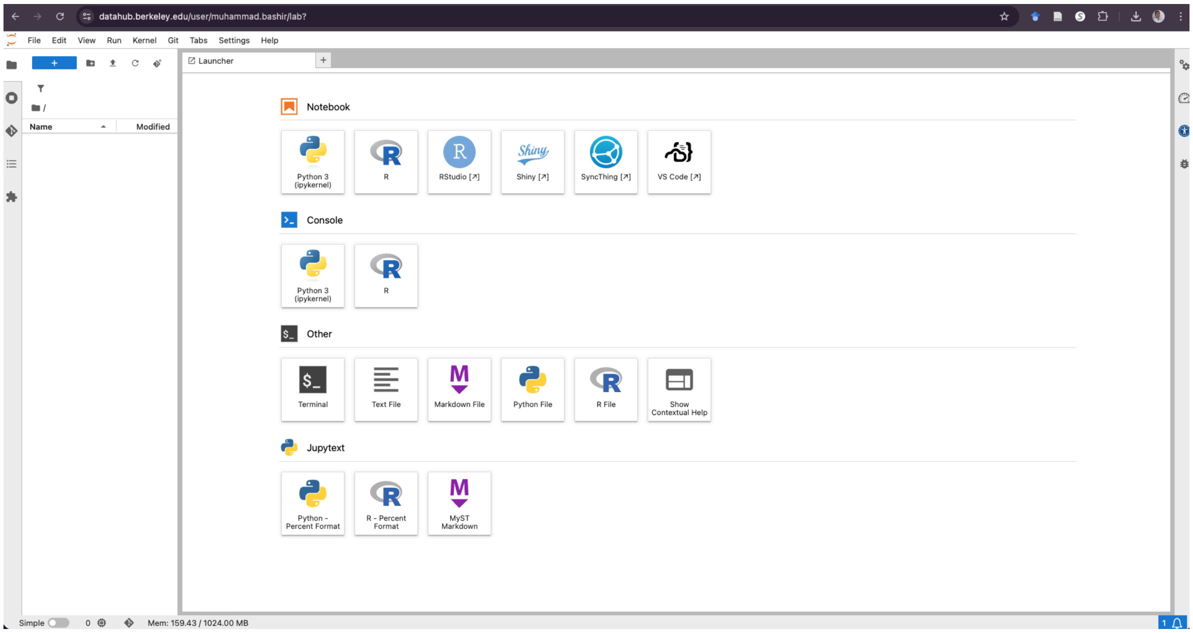
Why Datahub?

We highly recommend completing your coursework using **Datahub**.

- Datahub is a masterfully designed server accessible to everyone with a bCourses account (using your `username@berkeley.edu`).
- It comes fully configured for you to complete most tasks.
- Using Datahub allows you to avoid the "hard" process of trying to install and configure local copies of Python or R on your own machine.

How to Access Datahub

- **Python** is accessible at datahub.berkeley.edu.
- **R** is accessible at r.datahub.berkeley.edu.
- When you click a course assignment link, your browser will authenticate via bConnected, move you to Datahub, and launch a Jupyter notebook. This automatically creates a directory structure in your account and copies the files you need, so you never have to interface with GitHub directly.



When Datahub opens, you will typically see the launcher page shown above. Click on **Python 3 (ipykernel)** to open a Python notebook environment. You can upload data files there and use them in your work.

Getting Started in Python

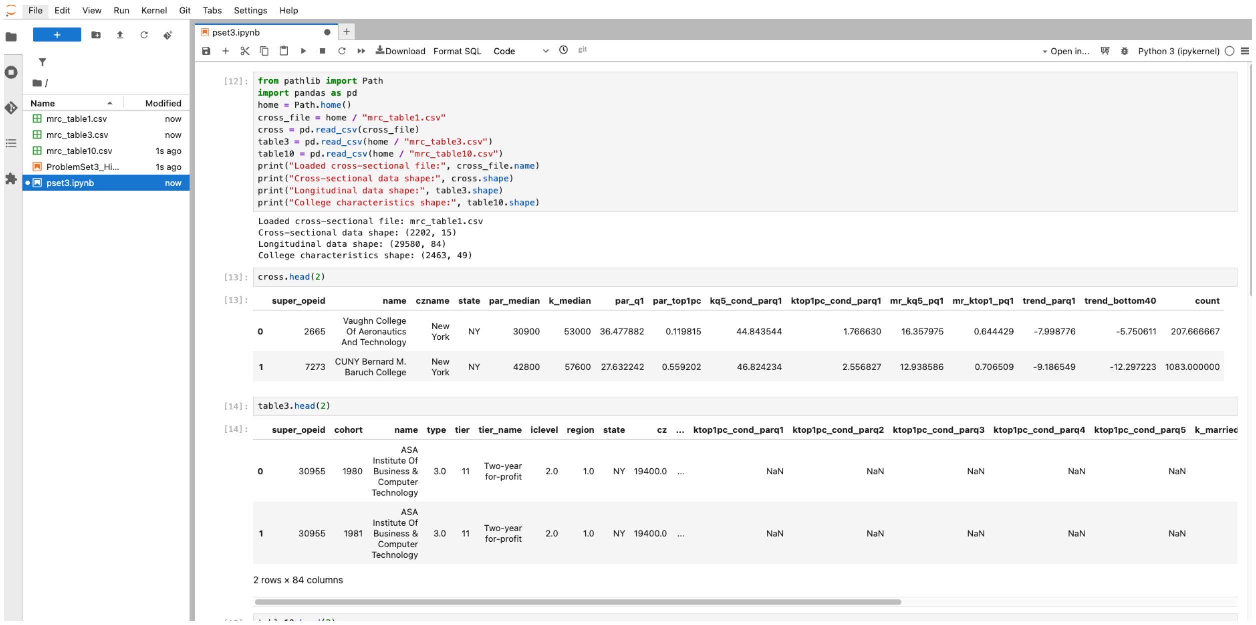
- After opening a Python notebook, upload the files `mrc_table1.csv` (or `mrc_table2.csv`), `mrc_table3.csv`, and `mrc_table10.csv` into your Datahub home directory.
- The code below uses `Path.home()`, which automatically points to your own Datahub home directory based on your username.
- Once those files are uploaded, you should be able to copy and paste this code directly.

Copy and paste the following into a Python notebook cell:

```
from pathlib import Path
import pandas as pd

home = Path.home()

cross_file = home / "mrc_table1.csv"
cross = pd.read_csv(cross_file)
table3 = pd.read_csv(home / "mrc_table3.csv")
table10 = pd.read_csv(home / "mrc_table10.csv")

print("Loaded cross-sectional file:", cross_file.name)
print("Cross-sectional data shape:", cross.shape)
print("Longitudinal data shape:", table3.shape)
print("College characteristics shape:", table10.shape)

cross.head()
table3.head()
table10.head()
```

This code loads the main files used in Problem Set 3 and displays the first few rows of each dataset.

Example of what your notebook may look like after you run the code:



Best Practices During the Semester

- **Save Frequently:** While Datahub typically auto-saves, it is good practice to manually save your work as well.
- **Watch Your Storage Limits:** Student accounts are typically limited to 1 GB of memory. Analyzing a large dataset might cause difficulties if you run out of space. You can check your storage size by opening a Terminal and executing `du -sh`. Be sure to delete unnecessary files.
- **Shared Directories:** If your coursework requires shared directories where instructors store large datasets, do not create a copy of those files in your home directory. Always read data directly from the shared directory.